

Core Pure Mathematics 2

Notes · Edexcel A Level Further Mathematics — Core Pure 2 · Chapter 1: Complex Numbers

Chapter 1: Complex Numbers

Why this matters

Core Pure 1 gave you the arithmetic of complex numbers — adding, multiplying, the Argand diagram, modulus and argument. Core Pure 2 takes that toolkit and uses it to unlock a genuinely new piece of mathematics: **de Moivre's theorem**, which turns powers and roots of complex numbers into simple trigonometric statements. From here you'll find roots of complex numbers (including the striking symmetry of roots of unity) and, later in the course, use these same ideas to derive trig identities. Everything below assumes you're completely fluent with modulus-argument form $z = r(\cos \theta + i \sin \theta)$; we won't re-derive that here.

Concept: de Moivre's theorem

For any complex number $z = r(\cos \theta + i \sin \theta)$ and any integer n :

$$z^n = r^n(\cos n\theta + i \sin n\theta)$$

This is proved by induction for positive integers n , and then extended to negative integers and to rational n (used for roots). The theorem says: to raise a complex number to a power, raise the modulus to that power and multiply the argument by that power. It is the single most useful fact in this chapter — almost everything else is an application of it.

Why the extension to negative and fractional n is safe. For a negative integer $n = -m$ ($m > 0$), de Moivre still holds because $z^{-m} = 1/z^m = 1/[r^m(\cos m\theta + i \sin m\theta)] = r^{-m}(\cos(-m\theta) + i \sin(-m\theta))$, using the fact that $1/(\cos \phi + i \sin \phi) = \cos(-\phi) + i \sin(-\phi)$ (multiply top and bottom by the conjugate, or note $\cos^2 \phi + \sin^2 \phi = 1$). So the same rule "multiply the argument, raise the modulus" just keeps working with a negative power. Fractional n is really the *roots* problem in disguise — that is exactly what the next section handles, and the reason there are n different answers (not one) is explained there.

Key vocabulary

Term	Meaning
Modulus-argument (polar) form	$z = r(\cos \theta + i \sin \theta)$, where $r = z $ and $\theta = \arg z$
Principal argument	The value of θ with $-\pi < \theta \leq \pi$
nth root of a complex number	Any w such that $w^n = z$
Roots of unity	The n solutions of $z^n = 1$

Primitive root of unity	A root of unity that is not a root of any smaller power $z^k = 1$ for $0 < k < n$
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Worked example 1 — Proving de Moivre's theorem by induction (n a positive integer)

WORKED EXAMPLE 1

Question. Prove that $(\cos \theta + i \sin \theta)^n = \cos n\theta + i \sin n\theta$ for all positive integers n .

Step 1. Base case $n = 1$: $(\cos \theta + i \sin \theta)^1 = \cos \theta + i \sin \theta = \cos(1 \cdot \theta) + i \sin(1 \cdot \theta)$. True. **(B1)**

Step 2. Assume true for $n = k$: $(\cos \theta + i \sin \theta)^k = \cos k\theta + i \sin k\theta$. **(M1) — stating the inductive hypothesis**

Step 3. Consider $n = k + 1$: $(\cos \theta + i \sin \theta)^{k+1} = (\cos \theta + i \sin \theta)^k (\cos \theta + i \sin \theta) = (\cos k\theta + i \sin k\theta)(\cos \theta + i \sin \theta)$ **(M1)**

Step 4. Expand and use the compound angle formulae: $= (\cos k\theta \cos \theta - \sin k\theta \sin \theta) + i(\sin k\theta \cos \theta + \cos k\theta \sin \theta)$ **(M1) — multiplying out and grouping real/imaginary parts** $= \cos(k\theta + \theta) + i \sin(k\theta + \theta)$ **(A1) — recognising the addition formulae** $= \cos((k+1)\theta) + i \sin((k+1)\theta)$ ✓

Step 5. Since it is true for $n = 1$, and true for $n = k$ implies true for $n = k+1$, by the principle of mathematical induction it is true for all positive integers n . **(A1) — correct induction conclusion**

Answer: de Moivre's theorem holds for all positive integers n . ■

EXAM TIP

It is tempting to "expand and hope" instead of clearly separating the five induction steps: base case, hypothesis, statement of what's being shown, the algebra, and the conclusion sentence. Learn the conclusion sentence word-for-word — Edexcel examiners specifically look for "true for $n = 1$, and true for $n = k \Rightarrow$ true for $n = k+1$, therefore true for all positive integers n by induction" as the final accuracy mark. Losing this one sentence costs an easy mark almost every time.

Concept: using de Moivre's theorem for calculation

Once proved, de Moivre's theorem is normally just *applied*, not re-derived. The main skill is converting a complex number into modulus-argument form, applying $z^n = r^n(\cos n\theta + i \sin n\theta)$, and converting back if needed.

Worked example 2 — Evaluating a power directly

WORKED EXAMPLE 2

Question. Find $(1 + i\sqrt{3})^6$, giving your answer in the form $a + bi$.

Step 1. Find modulus and argument: $r = \sqrt{1^2 + (\sqrt{3})^2} = \sqrt{4} = 2$, $\theta = \arctan(\sqrt{3}/1) = \pi/3$ (first quadrant). **(M1)** So $1 + i\sqrt{3} = 2(\cos \pi/3 + i \sin \pi/3)$.

Step 2. Apply de Moivre's theorem with $n = 6$: $(1 + i\sqrt{3})^6 = 2^6(\cos 6 \cdot \pi/3 + i \sin 6 \cdot \pi/3) = 64(\cos 2\pi + i \sin 2\pi)$ **(M1)(A1) — correct modulus 64 and correct angle 2π**

Step 3. Evaluate: $\cos 2\pi = 1$, $\sin 2\pi = 0$, so the answer is $64(1 + 0i) = 64$. **(A1)**

Answer: $(1 + i\sqrt{3})^6 = 64$.

Concept: nth roots of a complex number

To solve $w^n = z$, write z in modulus-argument form **using the general argument** $\theta + 2k\pi$ (not just the principal value), because the argument of a complex number is only defined up to multiples of 2π :

$$z = r(\cos(\theta + 2k\pi) + i \sin(\theta + 2k\pi)), k \in \mathbb{Z}$$

Then the n distinct n th roots are:

$$w = r^{1/n} [\cos((\theta + 2k\pi)/n) + i \sin((\theta + 2k\pi)/n)], \text{ for } k = 0, 1, 2, \dots, n-1$$

Taking k outside this range just repeats roots you already have, because adding 2π to θ before dividing by n adds $2\pi/n \cdot n = 2\pi$ to the angle — a full turn.

Key formula

KEY FORMULA

Task	Formula
Solving $z^n = w$ ($w \neq 1$)	$z = w ^{1/n} [\cos((\arg w + 2k\pi)/n) + i \sin((\arg w + 2k\pi)/n)], k = 0, \dots, n-1$
Solving $z^n = 1$ (roots of unity)	$z = \cos(2k\pi/n) + i \sin(2k\pi/n), k = 0, \dots, n-1$
Sum of all n th roots of unity	0 (for $n \geq 2$)
Product of all n th roots of unity	$(-1)^{n+1}$

Worked example 3 — Finding the cube roots of a complex number

WORKED EXAMPLE 3

Question. Find, in the form $r(\cos \theta + i \sin \theta)$ with $-\pi < \theta \leq \pi$, the three cube roots of $z = -8i$.

Step 1. Write $-8i$ in modulus-argument form: $r = 8$, and $-8i$ lies on the negative imaginary axis so $\theta = -\pi/2$. **(M1)** General form: $z = 8[\cos(-\pi/2 + 2k\pi) + i \sin(-\pi/2 + 2k\pi)]$.

Step 2. Take cube roots: modulus of each root is $8^{1/3} = 2$. Argument of each root is $(-\pi/2 + 2k\pi)/3$, for $k = 0, 1, 2$. **(M1)**

Step 3. Compute each argument: $k = 0$: $(-\pi/2)/3 = -\pi/6$ **(A1)** $k = 1$: $(-\pi/2 + 2\pi)/3 = (3\pi/2)/3 = \pi/2$ **(A1)** $k = 2$: $(-\pi/2 + 4\pi)/3 = (7\pi/2)/3 = 7\pi/6 \rightarrow$ subtract 2π to bring into range: $7\pi/6 - 2\pi = -5\pi/6$ **(A1)** — **correctly adjusting into $(-\pi, \pi]$**

Answer: The three cube roots are: $2(\cos(-\pi/6) + i \sin(-\pi/6))$, $2(\cos(\pi/2) + i \sin(\pi/2))$, $2(\cos(-5\pi/6) + i \sin(-5\pi/6))$

EXAM TIP

The most common error is forgetting the "+2kπ" entirely and only ever finding one root. Here is why it's needed: $\arg(z)$ is genuinely ambiguous up to full turns, and dividing that ambiguity by n is what spreads the n roots evenly around a circle. A good visual: draw the Argand diagram and show the n roots sit at the vertices of a regular n -gon, all at radius $r^{1/n}$. Once you see the picture, you stop forgetting roots.

Concept: roots of unity and their geometry

The n solutions of $z^n = 1$ are called the **nth roots of unity**. They are:

$$z = \cos(2k\pi/n) + i \sin(2k\pi/n), \quad k = 0, 1, \dots, n-1$$

Geometrically these are the n vertices of a regular n -gon inscribed in the unit circle, one vertex always at $z = 1$ ($k = 0$). A key algebraic fact: if ω is a primitive n th root of unity, then $1, \omega, \omega^2, \dots, \omega^{n-1}$ are *all* the n th roots of unity, and they satisfy

$$1 + \omega + \omega^2 + \dots + \omega^{n-1} = 0$$

since these are the roots of $z^n - 1 = 0$, and the coefficient of z^{n-1} is 0 (sum of roots = 0 by the factor theorem / Vieta's formulae). The same idea gives the product of all n roots: for $z^n - 1 = 0$, Vieta's formulae give product of roots = $(-1)^n \times (\text{constant term}) / (\text{coefficient of } z^n) = (-1)^n \times (-1) = (-1)^{n+1}$ — this is where the "Product of all n th roots of unity" row in the table above comes from.

A root of unity ω is called **primitive** if it isn't a root of any smaller power — i.e. the smallest positive integer m with $\omega^m = 1$ is $m = n$ itself. For $n = 4$, the four roots are $1, i, -1, -i$. Here i is primitive ($i^1=i, i^2=-1, i^3=-i, i^4=1$ — you need all 4 powers to get back to 1), but -1 is not primitive, since $(-1)^2 = 1$ already — it's a primitive **2nd** root of unity, not a 4th. This matters because *every* n th root of unity is some power of a primitive one, which is why the algebraic fact $1 + \omega + \omega^2 + \dots + \omega^{n-1} = 0$ is normally stated with ω taken to be primitive: only then does the list $\omega^0, \omega^1, \dots, \omega^{n-1}$ actually hit all n roots without repeats.

Worked example 4 — Roots of unity and a sum identity

WORKED EXAMPLE 4

Question. Write down the 5 fifth roots of unity, and hence show that $1 + \cos(2\pi/5) + \cos(4\pi/5) + \cos(6\pi/5) + \cos(8\pi/5) = 0$.

Step 1. The fifth roots of unity are $z_k = \cos(2k\pi/5) + i \sin(2k\pi/5)$, for $k = 0, 1, 2, 3, 4$. **(B1)**

Step 2. Since these are the 5 roots of $z^5 - 1 = 0$, and the coefficient of z^4 in $z^5 - 1$ is 0, the sum of the roots is 0: **(M1)** — using sum of roots = $-(\text{coefficient of } z^4)/(\text{coefficient of } z^5)$ $z_0 + z_1 + z_2 + z_3 + z_4 = 0$

Step 3. Write this out in real and imaginary parts: $[1 + \cos(2\pi/5) + \cos(4\pi/5) + \cos(6\pi/5) + \cos(8\pi/5)] + i[0 + \sin(2\pi/5) + \sin(4\pi/5) + \sin(6\pi/5) + \sin(8\pi/5)] = 0$ **(M1)**

Step 4. Since a complex number is zero only if both real and imaginary parts are zero, the real part gives: **(A1)** $1 + \cos(2\pi/5) + \cos(4\pi/5) + \cos(6\pi/5) + \cos(8\pi/5) = 0$ ■

Answer: as shown above — this follows directly from "sum of all n th roots of unity is zero".

(Since 5 is prime, every root here except $z_0 = 1$ is a primitive 5th root of unity.)

Chapter summary

Idea	Key point
de Moivre's theorem	$z^n = r^n(\cos n\theta + i \sin n\theta)$ for any integer n ; proved by induction for positive integers
Finding nth roots	Use general argument $\theta + 2k\pi$, then divide by n ; gives n distinct roots for $k = 0, \dots, n-1$
Roots of unity	Solutions of $z^n = 1$; evenly spaced around the unit circle, sum to 0
Geometric picture	n th roots of any z lie at the vertices of a regular n -gon of radius $ z ^{1/n}$

Common exam mistakes

- Forgetting the "+ $2k\pi$ " when finding roots, so only the principal root is found instead of all n roots.
- Using the wrong range of k (e.g. $k = 1$ to n instead of 0 to $n-1$), which can duplicate or omit roots.
- Giving final arguments outside $(-\pi, \pi]$ when the question asks for the principal range — always adjust by adding/subtracting 2π .
- Sloppy induction write-ups that skip the concluding sentence, losing the final accuracy mark even when the algebra is correct.
- Confusing "roots of unity" (solutions of $z^n = 1$) with " n th roots of z " for a general z — roots of unity are the special case $z = 1$.
- Mixing degrees and radians — Edexcel Further Maths expects radians throughout this chapter; a calculator left in degree mode will silently give wrong arguments.

You should now be able to

- Prove de Moivre's theorem for positive integers by induction, and use it (without re-proving it) for negative and fractional powers.
- Use de Moivre's theorem to evaluate powers of complex numbers in modulus-argument form.
- Find all n distinct n th roots of a complex number, expressing each in modulus-argument form within a given argument range.
- Find and use the n th roots of unity, including the geometric picture of a regular n -gon and the algebraic fact that they sum to zero.